

Title

How to Improve Software Using AI Tools

Abstract

Artificial Intelligence (AI) has become an important technology in modern software development. AI tools help developers write better code, detect bugs, automate testing, improve security, and increase overall software quality. This dummy research explores how AI tools enhance software development and concludes that AI improves quality and productivity when combined with human expertise.

Introduction

AI tools such as GitHub Copilot and ChatGPT assist developers in coding, debugging, testing, and documentation, enabling faster and more reliable software development.

Problem Statement

Many projects suffer from delays, coding errors, security vulnerabilities, and high maintenance costs due to manual processes and underuse of AI tools.

Objectives

1. Study AI in software development.
2. Identify quality improvements.
3. Examine benefits and limitations.
4. Suggest integration recommendations.

Research Questions

1. How do AI tools improve software development?
2. What are their benefits?
3. What challenges exist?
4. How can AI improve quality and maintenance?

Research Gap

Most previous studies focus on individual AI applications such as code generation or testing. Limited research examines the combined impact of multiple AI tools across the entire software development lifecycle while considering developer productivity and practical adoption.

Methodology

Qualitative literature review using journal articles, books, conference papers, and thematic analysis.

Expected Results

Improved code quality, fewer bugs, faster development, automated testing, better security, higher productivity, and lower maintenance costs.

Conclusion

AI is transforming software engineering by improving coding, testing, debugging, and maintenance. It complements human developers and enhances software quality.

References

Sommerville, I. Software Engineering.
Russell & Norvig. Artificial Intelligence: A Modern Approach.
IEEE articles on AI in Software Engineering.